Lecture Notes: Knowledge Graphs and Retrieval-Augmented Generation

Lecture notes · Coursework · Joselito Dizon · 2026-07-21

Notes on entity and relationship extraction, graph stores, embeddings, hybrid retrieval and grounding LLM answers in cited evidence.

Knowledge graphs

A knowledge graph stores entities as nodes and typed relationships as edges, which makes multi-hop questions answerable.

Typical relationship types in a document graph include cites, extends, implements, evaluates and supersedes.

Retrieval-augmented generation

Retrieval-augmented generation first retrieves relevant passages, then asks a language model to answer using only those passages.

Hybrid retrieval combines keyword scoring such as BM25 with embedding similarity, which helps with both exact terms and paraphrases.

Grounding means every claim in the answer can be traced to a retrieved passage; unsupported claims should be rejected, not hidden.

Evaluation

Retrieval quality is measured against a labelled ground truth, for example recall at k and the rate of unsupported claims.

Small local models can phrase answers well, but the facts must come from the retrieved evidence.